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# ExerPy: An open-source framework for automated exergy analysis

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**Software**

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## Summary

This paper presents ExerPy, a flexible and open-source software tool designed for automating the exergy analysis of thermodynamic systems. Exergy analysis offers a more comprehensive understanding of energy conversion processes by accounting for both the quantity and quality of energy, thereby revealing inefficiencies that traditional energy balance methods often overlook. ExerPy aims to reduce user effort and common errors by automatically defining exergy balances and integrating both physical and chemical exergy calculations. The tool enables engineers and researchers to perform in-depth analyses of a wide range of energy conversion systems, from simple cycles to complex power plants. It seamlessly interfaces with established simulation environments, including Aspen Plus, Ebsilon Professional, and TESPy, through a standardized JSON interface. ExerPy automatically identifies system components and defines exergy balances at both the component and system levels, thereby streamlining the analysis process. It outputs key metrics, such as exergy destruction and exergetic efficiency, which are essential for pinpointing sources of inefficiency and guiding optimization. The software's capabilities are validated using a combined-cycle power plant, where it effectively identifies inefficiencies in key components such as the combustion chamber and gas turbine system. By offering detailed insights into the performance of energy systems, ExerPy supports both academic research and industrial applications, facilitating more sustainable and resource-efficient energy conversion processes.

## Statement of need

Exergy analysis is an effective tool for evaluating and optimizing thermodynamic systems. Unlike conventional energy analysis, which accounts for the quantity of energy but neglects the quality and potential to provide useful work, exergy analysis provides deeper insights by considering both the quantity and quality of energy forms. This approach allows for a better understanding of where and why thermodynamic inefficiencies arise within a system. By quantifying irreversibilities that lead to exergy destruction and identifying their specific sources, engineers and researchers can develop strategies to improve efficiency, reduce costs, and support more sustainable energy conversion (Tsatsaronis, 1993). This makes exergy analysis a valuable tool not only for evaluating but also for designing and optimizing sustainable energy-conversion systems that prioritize sustainability and resource efficiency (Meyer et al., 2009; Petrakopoulou et al., 2017).

Despite its advantages, exergy analysis is not yet widely integrated into most commercial software used for thermodynamic assessments, which focus on energy and mass balance calculations. Simulation tools like Aspen Plus® and Ebsilon Professional® offer comprehensive

frameworks for modeling energy flows, but lack the built-in functionality needed to realize exergy analysis. This limitation reduces their effectiveness in evaluating system performance from a second-law perspective. The calculation of both physical and chemical exergy of material streams as well as an automated evaluation of the overall process was seamlessly integrated into the open-source software TESPy (Hofmann et al., 2022; Witte et al., 2022; Witte & Tuschy, 2020). While this was an important step in facilitating the application of exergy analysis, exergy analysis efforts still rely heavily on user input, are prone to incorrect interpretation of component balances, and lack interoperability with other open source or commercially available tools. These shortcomings have driven the demand for specialized, user-friendly, automated open-source software that enables exergy-based analyses and interoperates with both commercial and open-source tools.

To address these needs ExerPy provides a Python-based solution that automates exergy analysis of energy-conversion systems modeled in commercial software via a JSON data interface. The tool includes an API that automatically connects to Aspen Plus®, Ebsilon Professional®, or TESPy, autonomously identifies components and assigns exergy balances, enabling the detailed and accurate exergy analysis across the entire process. This level of automation not only streamlines the workflow, improving efficiency and accuracy, but also facilitates the identification of inefficiencies at both the system and component levels, thereby supporting the optimization of energy-conversion systems from an exergy perspective.

# Features

ExerPy is designed to perform exergy analysis through a structured workflow that integrates seamlessly with simulation tools. The initial implementation supports Ebsilon Professional®, Aspen Plus®, and TESPy. The framework is divided into two main modules: the data processing module, which manages the extraction and preparation of simulation data, and the exergy analysis module, which conducts the detailed exergy calculations. This modular design enhances flexibility and allows users to evaluate systems of varying complexity. The architecture is outlined in the following sections and is shown in Figure 1.

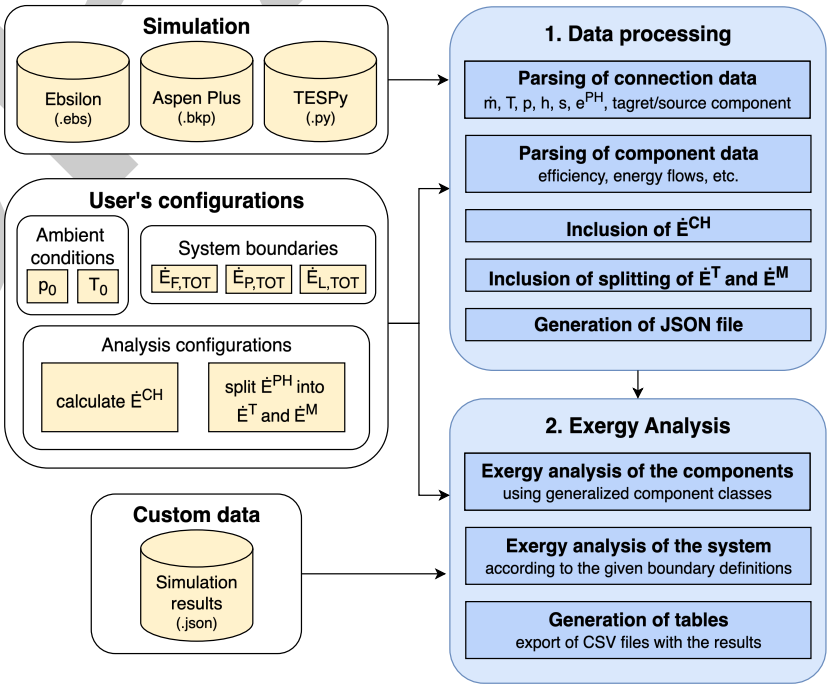


**Figure 1:** Structure of the ExerPy framework.

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## Data processing

The framework begins with the parsing of simulation data from models created in Ebsilon Professional®, Aspen Plus®, or TESPy, using the respective functions: `from_ebsilon`, `from_aspen`, and `from_tespy`. It is important to note that the physical exergy, calculated from the entropy and enthalpy of the streams, is parsed directly from the simulation tools. In the cases of `from_ebsilon` and `from_aspen`, the simulation is run and the necessary data are retrieved directly from the respective simulation environments. The integration with TESPy is slightly different: the simulation is performed beforehand within Python. The resulting TESPy simulation is then passed to ExerPy via the `from_tespy` function, which is responsible solely for calculating the exergy values. This allows updating the exergy values in case the ambient state is changed without rerunning the simulation. Independent of the simulation software, users also have the option to supply their own JSON file using `from_json`, which must conform to the required format for exergy analysis. In all cases, these functions instantiate an object of the `ExergyAnalysis` class, which internally calls the `_construct_components` function to build the system network.

During the parsing process, connection data such as mass flow ($m$), temperature ($T$), pressure ($p$), enthalpy ($h$), entropy ($s$), and physical exergy ($e^{\mathrm{PH}}$) are parsed, along with the identification of target and source components. In addition, component data including efficiency, energy flows, and other relevant thermodynamic properties are also extracted. Ambient conditions are taken directly from the simulation, or they can be specified by the user manually.

ExerPy also allows splitting the physical exergy into thermal ($e^{\mathrm{T}}$) and mechanical ($e^{\mathrm{M}}$) parts. This separation enables a more comprehensive analysis of thermodynamic processes, especially for components operating below ambient temperature (Morosuk & Tsatsaronis, 2019). These values are calculated using the native property functions of the simulation tools. In the initial release of ExerPy, this separation is not yet available in Aspen Plus due to limited access to thermodynamic functions, but it is planned for a future update.

After the data parsing is complete, ExerPy calculates the chemical exergy ($e^{\mathrm{CH}}$) of each material stream based on their molar or mass composition. The approach is adapted from TESPy (Hofmann et al., 2022) and the specific chemical exergy of each stream is calculated in ExerPy based on the thermodynamic reference environment developed by Ahrendts (Ahrendts, 1977). For pure substances, the specific chemical exergy is taken directly from tabulated data of the chosen thermodynamic model, while for mixtures like air or flue gas, the chemical exergy is calculated based on the values of individual components. This involves using molar fractions and assuming ideal behavior when the mixture is gaseous, with adjustments made for mixtures including condensable components (e.g., water) to account for both gas and liquid phases. The evaluation of chemical exergy is performed only if the user explicitly specifies it. However, it is important to note that its use becomes indispensable for systems involving chemical reactions or mixing processes, where chemical exergy variations are significant. Finally, all parsed and calculated data are consolidated into a standardized JSON file, independent of the simulation tool used, which includes all the necessary information for a comprehensive exergy analysis.

## Exergy Analysis

The framework performs exergy analysis at both the component and system levels. Each component of the system-such as turbines, compressors, and heat exchangers-is represented by a Python class that automatically assigns the exergy of the fuel and exergy of the product of the component. Using these definitions, ExerPy calculates relevant metrics for each component (i.e., the exergy destruction and exergetic efficiency). Thermal energy losses of components are included in their exergy destruction and streams discharged to the environment, are considered exergy losses of the overall system. This offers coherent calculations of inefficiencies of each individual component and enables targeted optimization. The class-based structure of the tool enables the user to define custom component classes specific to their problem. The component can then be registered to the ExerPy package using the `component_registry` decorator to

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be included in the automated evaluation. At the system level, the total exergy balance is determined by evaluating the exergy of streams crossing the system boundaries. To perform this analysis, and in the current release of the tool, it is necessary for the user to define the product, the fuel, and the exergy loss of the overall process. Further automation is planned for future versions of the platform. The system-level exergy analysis yields the overall exergetic efficiency and the total exergy destruction of the overall system. Finally, the framework offers the possibility to export the results as CSV files for further examination and integration into additional workflows.

# Use Case and validation

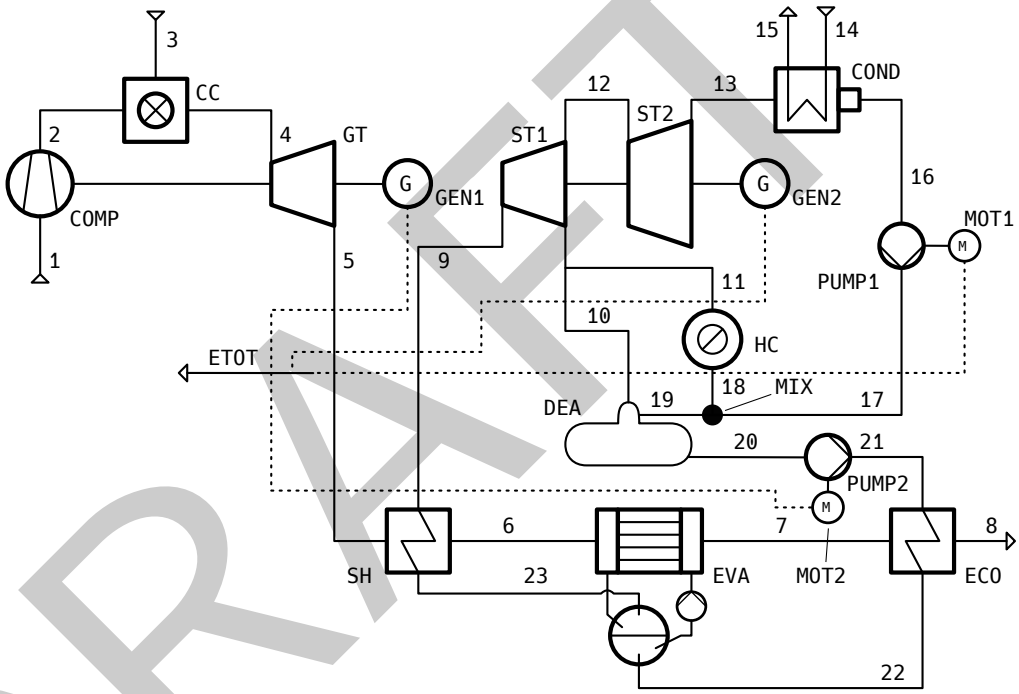


**Figure 2:** Flow sheet of the combined cycle power plant.

In this study, a combined-cycle power plant (CCPP) is analyzed to demonstrate the capabilities of the ExerPy framework for detailed exergy analysis. The analyzed CCPP, as illustrated in Figure 2, integrates a gas turbine system (GT) with a steam cycle to optimize energy conversion efficiency. The plant is designed to deliver 300 MW net electrical power, with 248 MW generated in the gas turbine, and 100 MW of thermal energy. The gas turbine system operates with a turbine inlet pressure of 15 bar and an inlet temperature of 1150 °C. The heat recovery steam generator (HRSG) generates steam at 50 bar and 505.6 °C, maintaining a minimum temperature difference of 25 K in the superheater. The steam cycle is split into high- and low-pressure stages. The high-pressure steam turbine expands the steam down to 15 bar. A portion of the expanded steam is extracted and condensed to provide heat, another portion is sent to the deaerator, and the rest passes through the low-pressure steam turbine. In the low-pressure steam turbine, the steam is further expanded and then condensed in the condenser (COND) using cooling water at ambient conditions. The electrical efficiency of the system is 48.5%. When coupled with the heat extraction process, the overall energetic efficiency of the system is 64.7%.

The results of the exergy analysis of the CCPP simulated with Ebsilon Professional® are shown

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in Table 1. The exergy destruction is calculated for all components of the system based on their exergy balance equation. Since the condenser is a dissipative component, its exergy product and fuel are not defined.

The combustion chamber (CC) is the component with the most significant exergy destruction and has the greatest impact on system performance. Together with the expander and the air compressor (COMP), the GT system accounts for approximately 88% of the exergy destruction in the overall plant.

**Table 1:** Exergy analysis results for each component of the combined-cycle power plant

| Component | $\dot{E}_\mathrm{F}$ [ MW ] | $\dot{E}_\mathrm{P}$ [ MW ] | $\dot{E}_\mathrm{D}$ [ MW ] | $\varepsilon$ [ % ] | $y$ [ % ] | $y^*$ [ % ] |
|---|---|---|---|---|---|---|
| CC | 632.23 | 436.46 | 195.77 | 69.04 | 30.52 | 76.98 |
| COMP | 241.98 | 231.04 | 10.95 | 95.48 | 1.71 | 4.30 |
| COND | – | – | 3.72 | – | 0.36 | 0.92 |
| DEA | 8.20 | 6.44 | 1.76 | 78.51 | 0.28 | 0.69 |
| ECO | 13.42 | 11.95 | 1.48 | 89.01 | 0.23 | 0.58 |
| EVA | 74.13 | 62.48 | 11.65 | 84.29 | 1.82 | 4.58 |
| GEN1 | 251.83 | 248.05 | 3.78 | 98.50 | 0.59 | 1.49 |
| GEN2 | 53.25 | 52.45 | 0.80 | 98.50 | 0.13 | 0.31 |
| GT | 509.78 | 493.81 | 15.97 | 96.87 | 2.49 | 6.28 |
| HC | 39.72 | 39.72 | 0.00 | 100.00 | 0.00 | 0.00 |
| MIX | 3.09 | 1.67 | 1.42 | 53.99 | 0.22 | 0.56 |
| MOT1 | 0.04 | 0.04 | 0.00 | 98.50 | 0.00 | 0.00 |
| MOT2 | 0.46 | 0.46 | 0.01 | 98.50 | 0.00 | 0.00 |
| PUMP1 | 0.04 | 0.03 | 0.01 | 81.17 | 0.00 | 0.00 |
| PUMP2 | 0.46 | 0.40 | 0.06 | 87.30 | 0.01 | 0.02 |
| SH | 33.35 | 29.01 | 4.35 | 86.97 | 0.68 | 1.71 |
| ST1 | 28.46 | 27.47 | 1.00 | 96.50 | 0.16 | 0.39 |
| ST2 | 28.78 | 25.78 | 3.00 | 89.58 | 0.47 | 1.18 |
| TOT | 641.45 | 339.72 | 254.32 | 52.96 | 39.65 | 100.00 |

The following specifications are used in the exergy analysis of the overall plant:

$$\dot{E}_\mathrm{F} = \dot{E}_1 + \dot{E}_3 \tag{1}$$

$$\dot{E}_\mathrm{P} = \dot{W}_\mathrm{el,net} + \dot{E}_\mathrm{q} \tag{2}$$

$$\dot{E}_\mathrm{L} = \dot{E}_8 + \dot{E}_{15} - \dot{E}_{14} \tag{3}$$

where, $\dot{W}_\mathrm{el,net}$ is the net electric power and $\dot{E}_\mathrm{q}$ is the exergy related to the extracted heat flow rate. The value of $\dot{E}_\mathrm{q}$ is determined as the difference between the thermal exergy of the inlet and outlet streams. The temperature level at which the heat is transferred is not specified since the cold stream is not simulated in this example.

The system results in an exergetic efficiency of approximately 53%, which is comparable to values reported in the literature (Ameri et al., 2008; Ersayin & Ozgener, 2015; Petrakopoulou et al., 2012). The exergy product amounts to around 340 MW, with 300 MW attributed to electric power and the remaining 40 MW to thermal energy. The exergy analysis of the entire system shows that the exergy of the product is considerably lower than the energy of the product due to the low exergy value of the heat output (25.0%). This underscores the significance of incorporating exergy calculations when assessing and comparing product of different quality.

The results of the exergy analysis of the CCPP simulated with Aspen Plus® and with TESPy show a maximum difference of 1% compared to the simulation results from Ebsilon Professional®, validating the accuracy and confirming the flexibility of the tool. Additional

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applications of ExerPy, such as the CGAM process (Valero et al., 1994) and a heat pump, are also provided in the GitHub repository.

# Nomenclature

| Abbr. | Description | Abbr. | Description |
|---|---|---|---|
| CC | Combustion chamber | CCPP | Combined cycle power plant |
| COMP | Compressor | COND | Condenser |
| DEA | Deaerator | ECO | Economizer |
| EVA | Evaporator | GEN | Generator |
| GT | Gas turbine expander | HC | Heat consumer |
| MIX | Mixer | PUMP | Pump |
| SH | Superheater | ST | Steam turbine |
| TESPy | Tool for energy system simulation in Python | | |

| Symbol | Meaning | Symbol | Meaning |
|---|---|---|---|
| 0 | Ambient state | CH | Chemical (exergy) |
| D | Destruction | F | Fuel |
| i | Component index | L | Loss |
| M | Mechanical (exergy) | P | Product |
| PH | Physical (exergy) | T | Thermal (exergy) |

| Symbol | Meaning | Symbol | Meaning |
|---|---|---|---|
| $\dot{E}$ | Exergy rate [MW] | $h$ | Enthalpy [kJ/kg] |
| $m$ | Mass flow rate [kg/s] | $p$ | Pressure [bar] |
| $s$ | Entropy [kJ/(kg · K)] | $T$ | Temperature [°C] |
| $\dot{W}$ | Power rate [MW] | $y$, $y^*$ | Exergy yield parameters [%] |

| Symbol | Meaning | Symbol | Meaning |
|---|---|---|---|
| $\varepsilon$ | Exergetic efficiency [%] | $\eta$ | Energetic efficiency [%] |

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